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ETW3482

Week 8 Tutorial



This week's tutorial will have **two sessions**

1. **Questions for revision.**
2. **Hands-on practical session: Support Vector Machine**



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What does a Support Vector Machine use to classify the data?

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To train a support vector machine is to obtain the hyperplane's _____, w and _____, b, which can best separate the observations with different targets in the training data.

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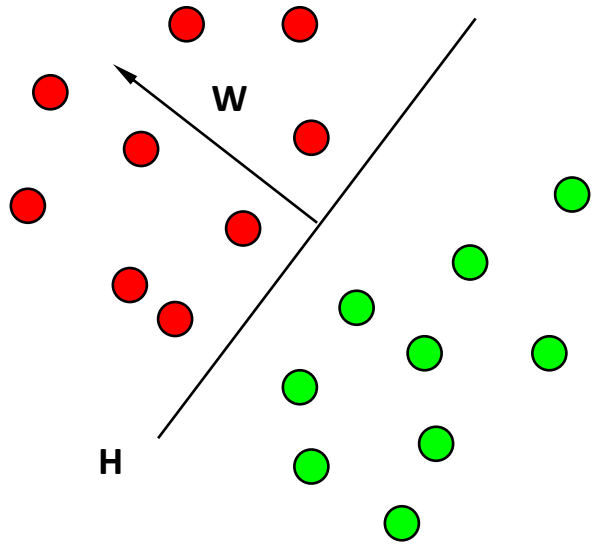
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Linear Separation of the Training Data

- A separating hyperplane H is given by the following:
  - the normal vector w
  - an additional parameter, b, called bias

$$H = \{\langle w, x \rangle + b = 0\}$$

Dot product



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Which of the following is TRUE for the prediction of Support Vector Machines (SVM)?

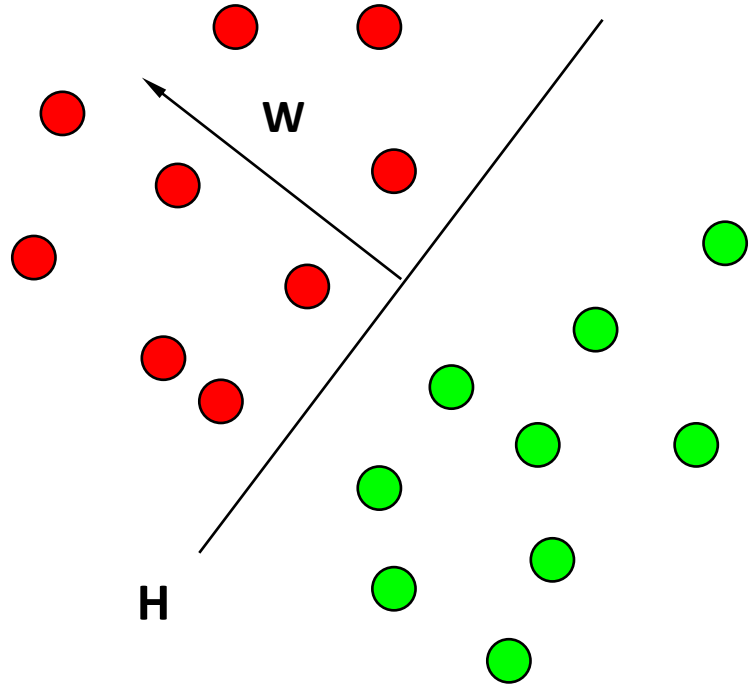
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Training versus Prediction

- Training:
- Select w and b in such a way that the hyperplane separates the training data – that is, the construction of a hyperplane.

- Prediction of the class for a new observation:
  On which side of the hyperplane (+/-) is the new data point located?



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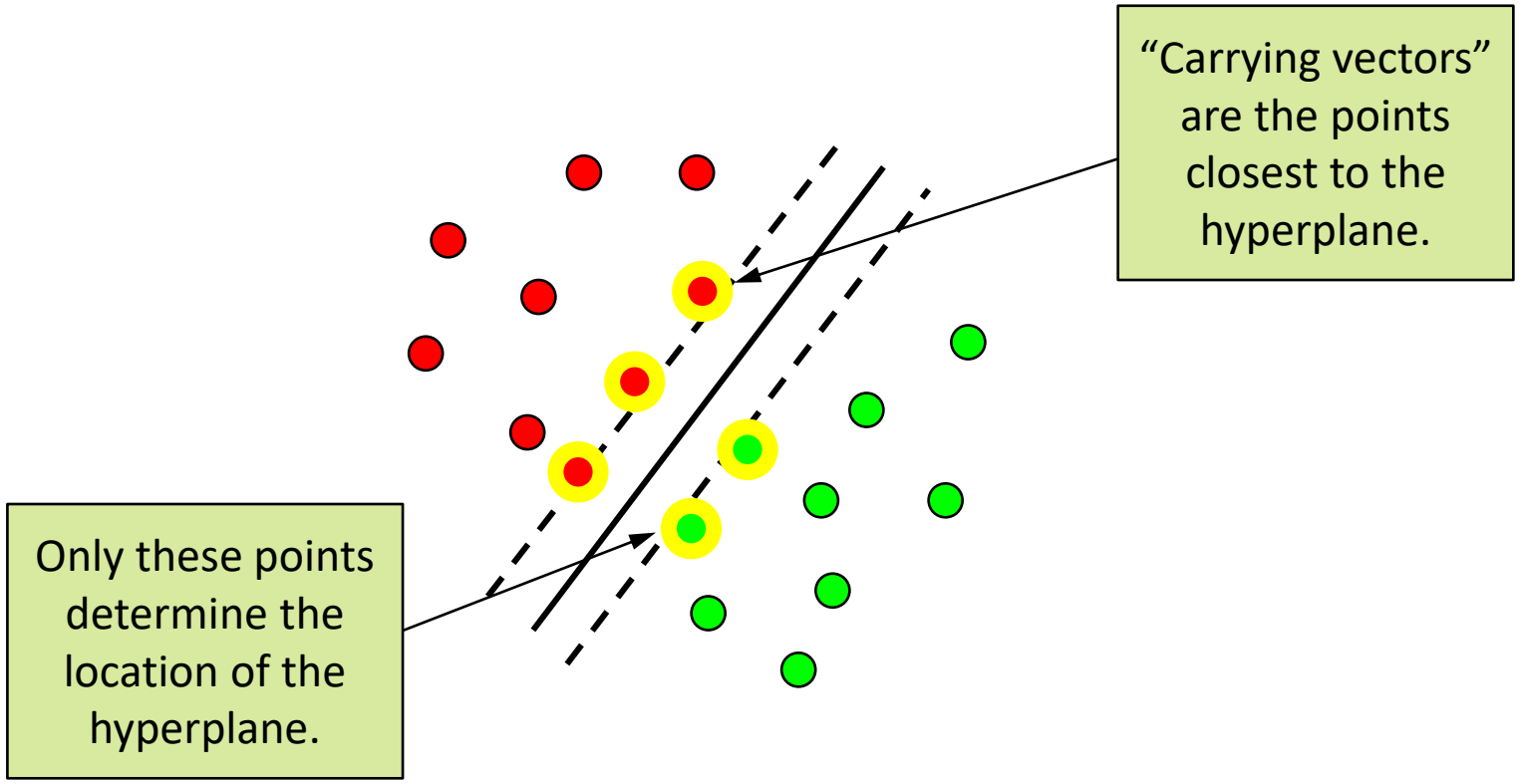


Which observations are support vectors that determine the location of the maximum-margin hyperplane?

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What Are the Support Vectors?



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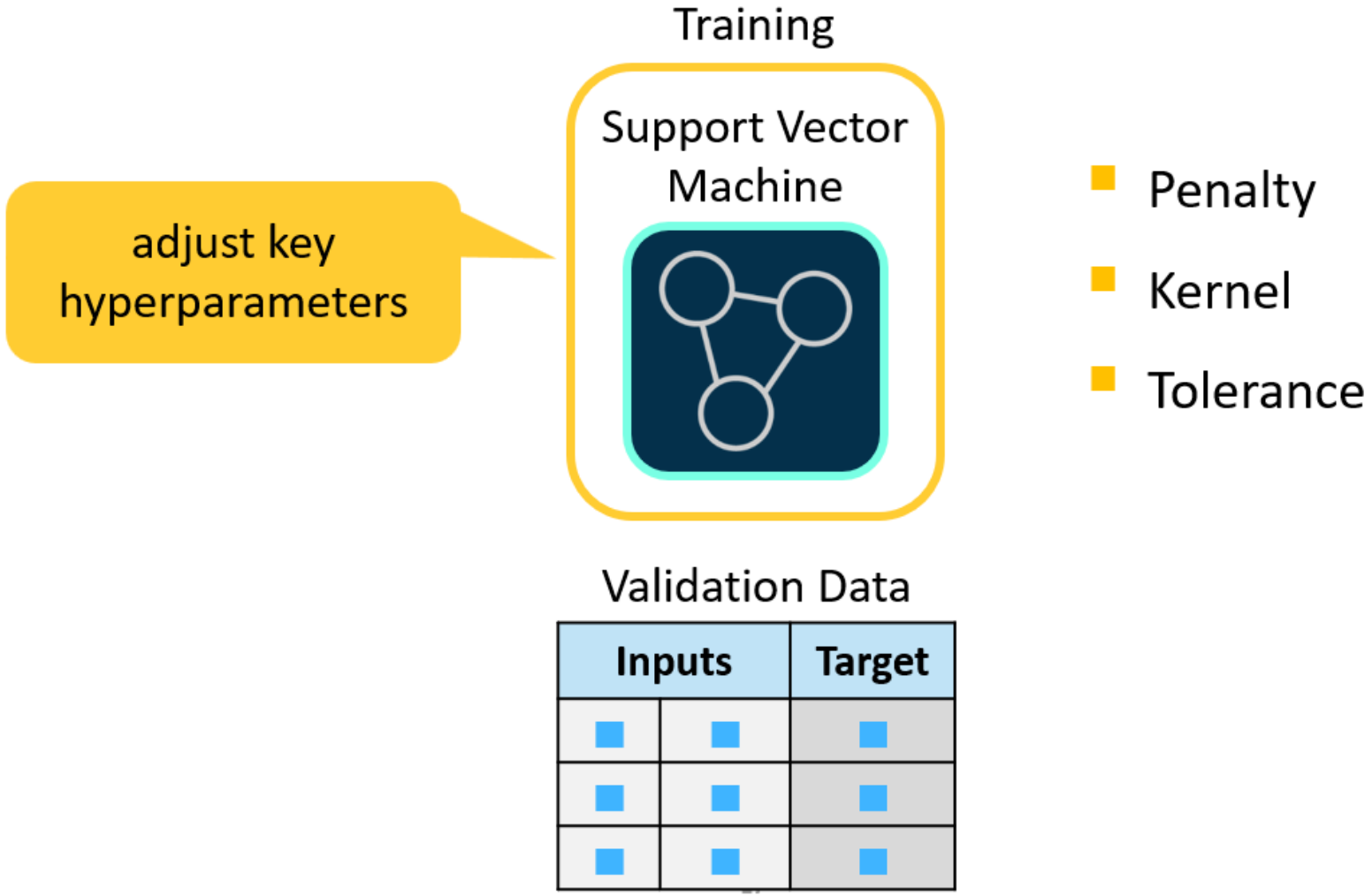


Which of the following is TRUE for optimising the complexity of SVMs?

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Improving the Support Vector Machine Model



- Penalty
- Kernel
- Tolerance

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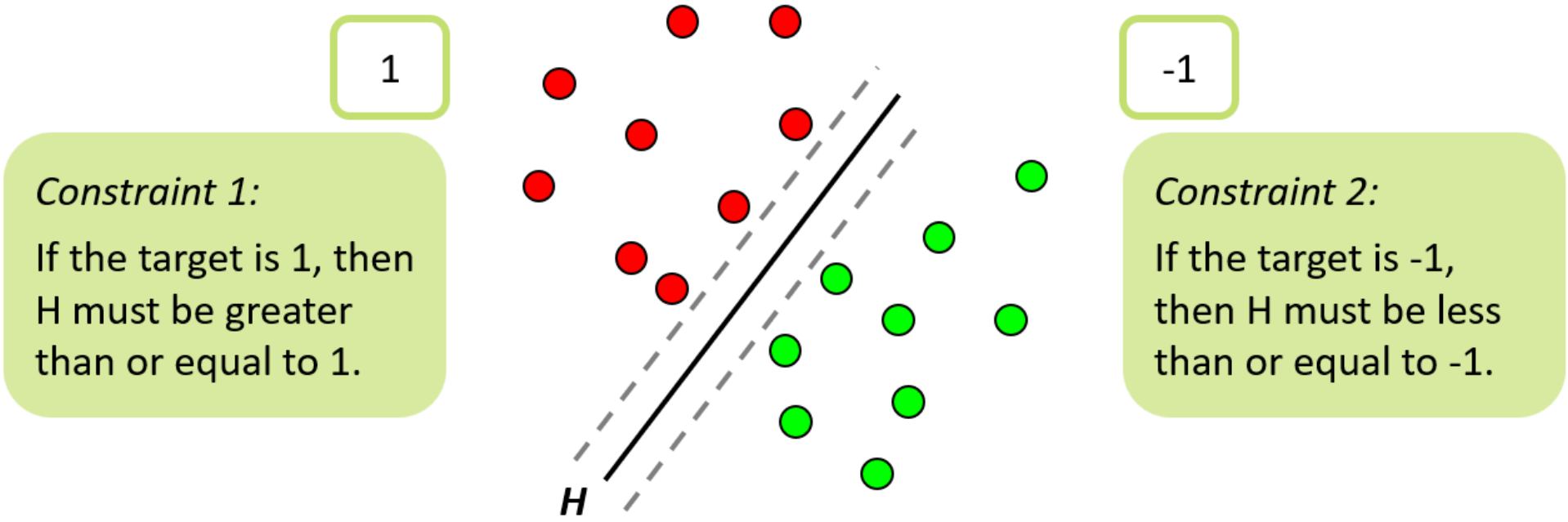
A Support Vector Machine is used to classify the binary target data. The red observations are labelled as 1, as shown in the diagram. What should we label for the green observations?

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Optimisation Problem

The solution for finding the separating hyperplane H becomes an *optimisation* problem under two constraints:

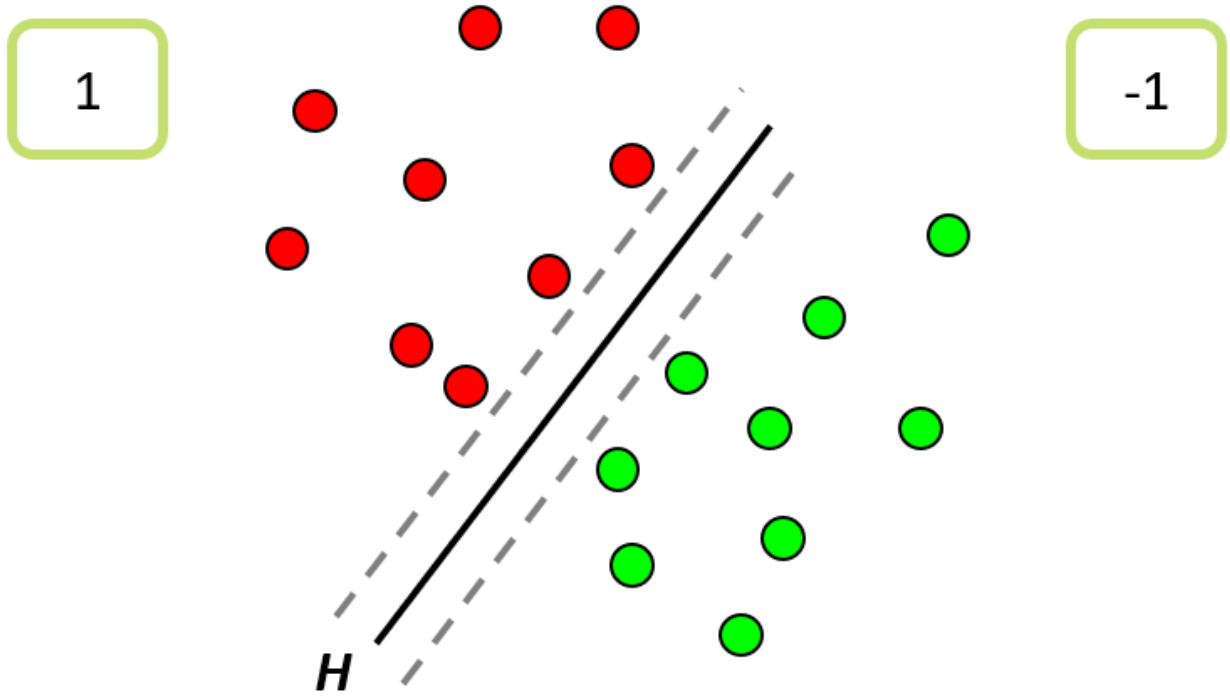


The binary target is written as +/- 1 for mathematical convenience.

Optimisation Problem

The two constraints can be combined into a single constraint:

The product of the target and H must be greater than or equal to 1 for all cases.



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The optimisation process to obtain the estimates of the normal vector, w and bias parameter, b, is based on a constraint, $y\cdot(\langle w,x\rangle+b)\geq 1$. Which of the following constraints is this constraint obtained from?

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Optimisation Problem

- Mathematically, these constraints are written as follows:

$$\langle w, x_i \rangle + b \geq 1 \quad \text{if } y_i = 1 \text{ and}$$
$$\langle w, x_i \rangle + b \leq -1 \quad \text{if } y_i = -1$$

- However, these two constraints can be combined into a single constraint where

$$y_i \cdot (\langle w, x_i \rangle + b) \geq 1 \quad \text{for } i = 1, 2, 3, \ldots, n.$$

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A (a)_ incorporates the distance between an error point and the hyperplane and the error weight or regularisation parameter, C, in the estimation process to accommodate data that is (b)_.

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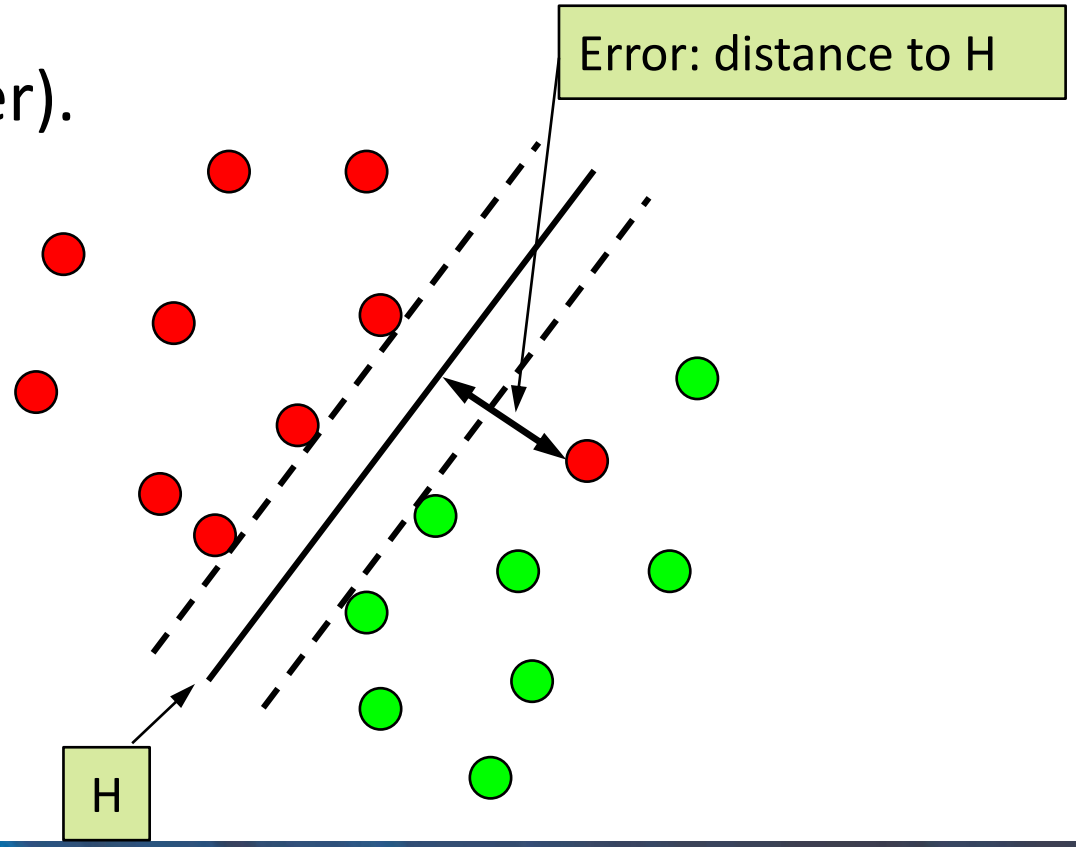
Training Data Not Linearly Separable

Errors are accounted for during the optimisation process:

Penalty: C*(distance to hyperplane)

C is an error weight (regularization parameter).

We have a so-called **soft margin hyperplane** if the data points are not linearly separable. In this case, we need to account for errors that the separating hyperplane might make. During optimisation, the distance between a point in error and the hyperplane is typically denoted by ξ.



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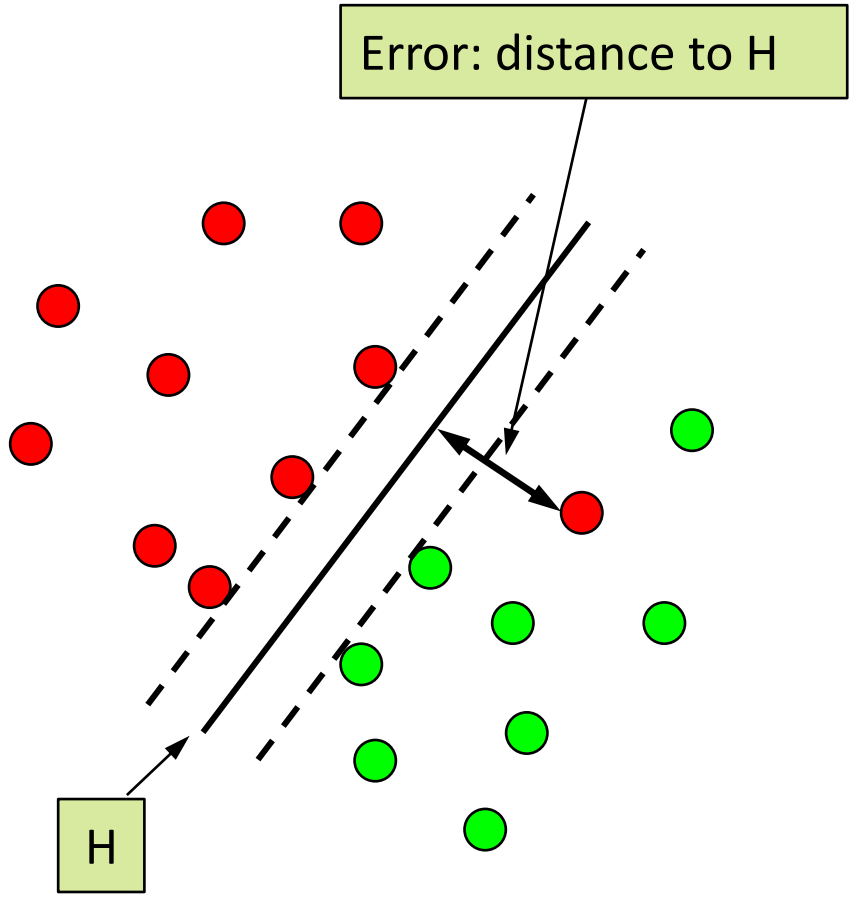
When the regularisation parameter, C is (a)_, the optimisation is focused on maximising the margin, which allows for more classification mistakes. Therefore, a larger C value will tend to (b)_ the training data and be more sensitive to outliers.

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Training Data Not Linearly Separable

- For large values of C, the optimization chooses a smaller-margin hyperplane if that hyperplane does a better job of getting all the training points classified correctly.
- Conversely, a very small value of C causes the optimizer to look for a larger-margin separating hyperplane, even if that hyperplane misclassifies more points. For very tiny values of C, you should get misclassified examples, often even if your training data is linearly separable.



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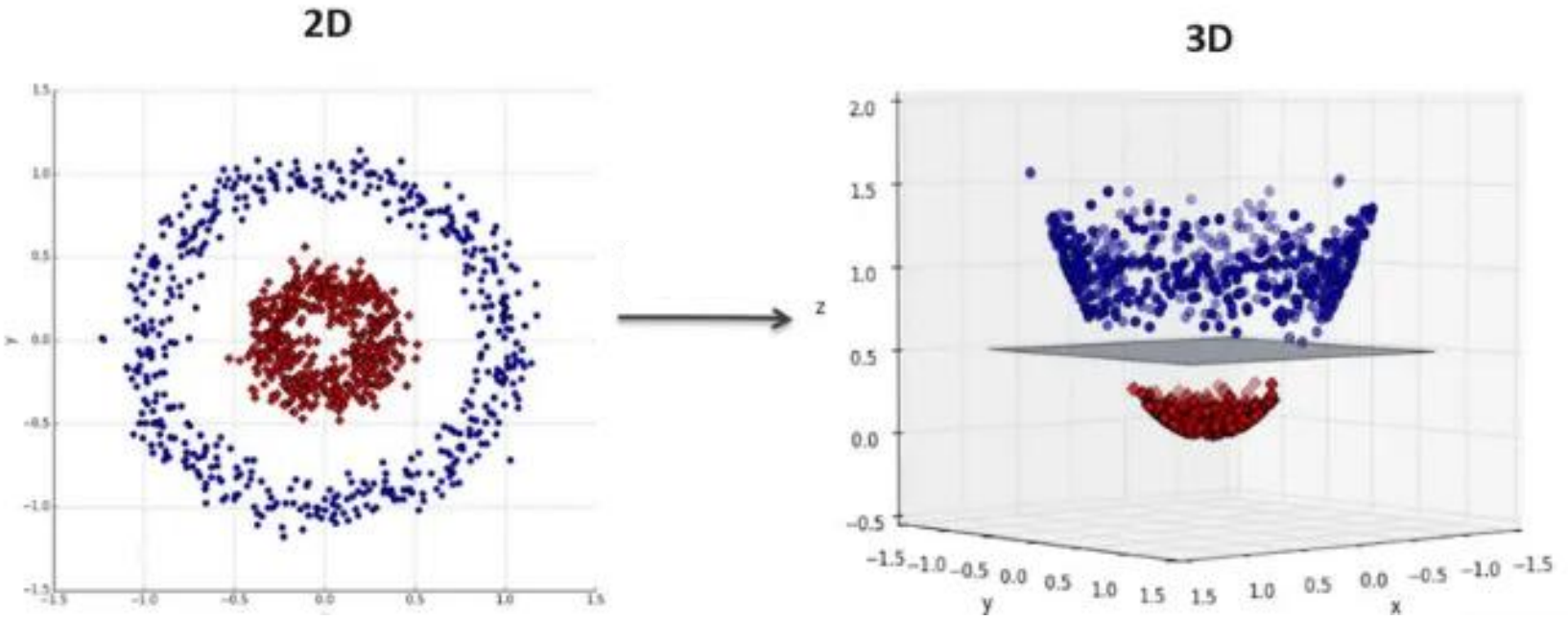


When the data is _____, it can be projected into _____, where it may become linearly separable. The _____ can be used to transform the inputs into a new feature without performing the dot product on the transformed data.

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Problem: Not Linearly Separable Data Points



Important Observations

- Estimating the parameters for a support vector machine and classifying a new case both require dot products of the form:

$$\langle x_i, x_j \rangle$$

- The solution in a higher dimension requires dot products on transformed data:

$$\langle \Phi(x_i), \Phi(x_j) \rangle$$

- **Problem:**
  Dot Products of the form $\langle \Phi(x_i), \Phi(x_j) \rangle$ are difficult to calculate.

Solution: The Kernel Trick

- We use a kernel function, living in the input space, but behaving as a dot product in the feature space:

$$K(x_i, x_j) = \langle \Phi(x_i), \Phi(x_j) \rangle$$

- Trick: We do not have to know $\Phi(x)$ explicitly!

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Which of the following are the options for the kernel function in the SAS Model Studio's SVM node?

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Which of the following hyperparameters can be altered for the complexity optimisation process to get different SVM models?

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Parameters for SVMs for Classification

- The kernel function and its parameters
  - Avoids the explicit mapping that is needed to get linear learning algorithms to learn a nonlinear function or decision boundary.
- The penalty C (regularisation term)
  - Balances model complexity and training error. A larger penalty value creates a more robust model at the risk of overfitting the training data.
- The tolerance
  - Balances the number of support vectors and model accuracy. A tolerance value that is too large creates too few support vectors, and a too small value overfits the training data.

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The Support Vector Machines can be used to identify the factors that are significant to the target.

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PART 2:
PRACTICAL HANDS-ON SESSION



Preparation for the practical session

1. Open the demo project.
2. Use the CPML demo pipeline template to create a new pipeline.
3. Name the new pipeline as the Week 8 tutorial.
4. Add an SVM node after the variable selection node.
5. Refer to the SVM node property panel.

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Which of the following is the main purpose of the scale inputs option?

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What is the purpose of "Use missing" option?

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Practical task 1

1) Add another SVM node after the variable selection node.
2) Name the SVM node as Overfitting SVM.
3) Change the value of Penalty to 100.
4) Change the value of Tolerance to 0.01.
5) Run the Overfitting SVM node.

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What is the average square error and misclassification rate on the validation data of the Overfitting SVM?

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Practical task 2

1) Add another SVM node after the variable selection node.
2) Name the SVM node as Underfitting SVM.
3) Change the value of Penalty to 0.001.
4) Change the value of Tolerance to 0.01.
5) Run the Underfitting SVM node.

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What is the average square error and misclassification rate on the validation data of the Underfitting SVM?

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Practical task 3

1) Add another SVM node after the variable selection node.
2) Name the SVM node as Polynomial SVM.
3) Change the Kernel to Polynomial and Polynomial degree to 2.
4) Change the value of Tolerance to 0.01.
5) Run the Polynomial SVM node.

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What is the average square error and misclassification rate on the validation data of the Polynomial SVM?

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Practical task 4

1) Run the Pipeline.
2) Refer to the Model Comparison node result.
3) Then answer the following question.

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Which model is the champion model?

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WRAP UP



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Q&A



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Audience Q&A

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